

Saksham Sharma

Delhi, India | +91 7626925956 | [Email](#) | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

Education

Thapar Institute of Engineering and Technology
Bachelor of Technology in Electronics and Computer Engineering

Patiala, Punjab
Aug 2023 – May 2027

Projects

URL Shortener API with Click Analytics

July 2026

Python, FastAPI, PostgreSQL (Neon), SQLAlchemy, pytest, Alembic, async/await

- Engineered production-grade REST API with collision-free short code generation (Base62 bijection, 56 billion combinations), real-time click analytics, and async SQLAlchemy; deployed on Railway with PostgreSQL (Neon). Fixed route registration bug where catch-all routes shadowed specific endpoints.
- Built comprehensive test suite (13 tests, 100% passing) with pytest + pytest-asyncio; debugged async event loop issue affecting SQLAlchemy connection pooling. Discovered and fixed .gitignore UTF-16 encoding (silently breaking Git patterns) and rotated credentials post-security discovery.
- Designed normalized PostgreSQL schema with indexed columns (O(1) lookups), foreign key constraints (ON DELETE CASCADE), and Alembic migrations for reproducible deployments. Prevented SQL injection via parameterized queries; removed manual create_all() from application lifespan.

Mini ATS - Candidate Pipeline Tool

July 2026

React, TypeScript, Tailwind CSS, ASP.NET Core, Entity Framework Core, SQLite, Docker, Railway

- Built and deployed full-stack applicant tracking system with job posting, PDF/DOCX resume upload, and recruiter dashboard for 5-stage pipeline (Applied to Rejected/Offer). Frontend deployed on Vercel, backend on Railway with Docker; cross-origin requests via CORS.
- Designed 5-endpoint REST API in ASP.NET Core (Minimal API) with EF Core, normalized Job-Candidate relational model, cascading status updates, and multipart form handling for file uploads. Implemented client/server-side validation with proper error responses.
- Developed responsive React UI with TypeScript for type safety and Tailwind CSS styling. Made deliberate architectural tradeoffs (SQLite for rapid iteration, Minimal API over MVC) to prioritize a fully working, well-documented system over unfinished complexity.

Token Bucket Rate Limiter Service

July 2026

Python, Flask, Threading, Concurrency Control, Lock Management

- Built standalone, networked rate-limiting API implementing token bucket and sliding window algorithms from first principles with configurable algorithms and standard HTTP rate-limit headers (RateLimit-Limit, RateLimit-Remaining, RateLimit-Reset).
- Designed thread-safe per-client bucket locking strategy using per-instance locks guarded by a shared map lock, guaranteeing zero double-spending under concurrent access. Verified correctness with load test firing 150 concurrent requests via thread pool.
- Implemented admin endpoint for per-client configuration (rate, burst size, algorithm mode); demonstrated mastery of synchronization primitives, lock contention analysis, and production-grade concurrency patterns.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, SQL, HTML5, CSS3

Backend: FastAPI, Flask, ASP.NET Core, Node.js, Express, REST APIs, async/await

Databases: PostgreSQL, SQLite, SQLAlchemy, Entity Framework Core

Frontend: React, Next.js, TypeScript, Tailwind CSS

Tools: Docker, Railway, Vercel, Git, pytest, pytest-asyncio, Alembic

Core CS: Data Structures, Algorithms, OOP, Concurrency, System Design, Security

Leadership and Extracurricular

Vice President, TED-Ed Club - Spring Dale Senior School (1-year term)

Participant, Smart India Hackathon (SIH)